SYNTHETIC DATA — 교육용 가상 데이터이며 실제 제품 사양·운영 기준과 무관합니다.

XGB PLC CPU 16 I/O (PLC-XGB-XBM-DR16S) — Datasheet

1. Overview

Compact PLC CPU, 8 DC inputs / 8 relay outputs, modular expansion. Product family: Programmable logic controllers. Lifecycle status: **ACTIVE**.

2. Technical specifications

2.1. Ratings

Item	Value
Built-in I/O	8 DC inputs (24 V) / 8 relay outputs
Max expansion modules	7 modules
Relay output rating	AC 250 V / DC 24 V, 2 A per point

2.2. Performance and characteristics

Item	Value
Program capacity	20 k steps
Processing speed	0.16 µs/step (basic instruction)

2.3. Endurance and communication

Item	Value
Built-in communication	RS-232C 1 ch + RS-485 1 ch

2.4. Protection and options

Item	Value
Backup battery life	Approx. 5 years (lithium)

3. Installation data

3.1. Terminals

- Dimensions (W x H x D): 30 x 90 x 64 mm
- Weight: 0.3 kg
- Degree of protection: IP20
- Operating temperature: 0 to +55 °C

4. Standards and certifications

Applicable standards:

- IEC 61131-2

Certifications held:

- CE
- UL
- KC

Certificates and declarations of conformity are available from the Certification & Compliance team on request. Certifications not listed above are not held for this product; project-specific evaluation can be requested.

5. Ordering information

Item	Value
Ordering code	PLC-XGB-XBM-DR16S
Standard lead time	3 weeks ex-works
Minimum order quantity	1
Warranty	24 months (see POL-WARRANTY)
Firmware	latest 3.20 (see RN-PLC-XGB-XBM-DR16S)

6. Related documents

- IG-PLC-XGB-XBM-DR16S — Installation and Commissioning Guide
- TG-PLC-XGB-XBM-DR16S — Troubleshooting and Error Code Guide
- RN-PLC-XGB-XBM-DR16S — Firmware Release Notes
- FAQ-PLC — Programmable logic controllers FAQ

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XGB PLC CPU 16 I/O (PLC-XGB-XBM-DR16S) — Installation and Commissioning Guide

1. Safety

Installation, wiring and commissioning must be performed by qualified personnel. Disconnect and lock out all power sources before work. Verify absence of voltage.

2. Receiving and storage

Inspect the packaging on arrival and report damage within 7 days (see POL-RETURN). Store indoors in a dry place between -25 and +55 °C, relative humidity below 85% without condensation.

3. Mounting and clearances

- Required clearances: 50 mm above and below the base for ventilation; keep 100 mm from power cables
- Operating temperature: 0 to +55 °C
- Degree of protection: IP20
- Above 1,000 m altitude, apply derating per the datasheet.

4. Wiring and tightening torques

- Terminal tightening torque: see dimension drawing
- Use copper conductors rated for at least 75 °C.
- Separate power and signal cables by at least 200 mm.

5. Commissioning procedure

1. Check power supply and base grounding (FG)
2. Download I/O parameter and program
3. Verify each input and output point
4. Check communication settings
5. Back up the final project

6. Pre-energisation checklist

☐ Product rating matches the drawing
☐ All terminals tightened to specified torque
☐ Protective earth connected
☐ Insulation resistance measured and recorded
☐ Settings / parameters recorded

7. Common commissioning issues

Symptom	Likely cause	See
CPU stops with watchdog error	Scan time exceeds watchdog due to program loop	TG-PLC-XGB-XBM-DR16S §3.1
Random watchdog error in noisy environment	Electrical noise due to improper grounding	TG-PLC-XGB-XBM-DR16S §3.2
I/O configuration mismatch after module replacement	Replacement module type differs from I/O parameter	TG-PLC-XGB-XBM-DR16S §3.3
Battery low warning	Backup battery end of life	TG-PLC-XGB-XBM-DR16S §3.4

On-site commissioning support can be requested through the Global CS Center. For programmable logic controllers the request is handled by the technical support team.

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XGB PLC CPU 16 I/O (PLC-XGB-XBM-DR16S) — Firmware Release Notes

1. Summary

Latest firmware: **3.20**. Updating to the latest version is recommended for all units.

2. Version history

2.1. Version 3.20 (2025-11-03)

- Added: Modbus RTU master up to 64 blocks
- Fixed: RTC drift > 30 s/month at high ambient temperature

2.2. Version 3.10 (2025-02-17)

- Fixed: P.04 program checksum error after repeated power cycling within 1 s

2.3. Version 3.00 (2024-07-01)

- Baseline release

3. Update procedure

1. Back up parameters / project before updating.
2. Connect the engineering tool (v4.6 or later).
3. Do not switch off power during the update.
4. After the update, verify the version and restore parameters if required.
5. If the update fails, do not power-cycle the unit repeatedly; contact support for the recovery procedure.

4. Compatibility

Parameter files from older versions can be loaded; files from newer versions cannot be loaded into older firmware.